

CELLGUARD · DESIGN DECISIONS REGISTER

CellGuard design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cellguard		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out: switches, status light and light pipe priced; value engineering restated (USD 159.15, USD 19.15 over the target); To confirm items for the switches, status light and light pipe
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 12 (CGD-DDR-003 accepted); moved to decisions made; To confirm item 5 and 8 and the value-engineering savings updated
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CellGuard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	A protector fuse rated for 40 A continuous at 60 V DC or more, and its price	R4 depends on it; the estimate is already over the value-engineering target, and this price could add to it	CGD-CAL-001 section 4; BOM line 14
2	The fuse holder: base no larger than 44 x 32 mm, four M4 fixing holes at the drawn positions, M8 terminal studs, and fuse blade faces 12.6 mm above its mounting face	The copper link is flat only if the blade faces and the B+ stud shoulder are at the same height; the plate holes follow the holder	CGD-DDR-003 P5; BOM line 8
3	The fuse and holder mass	The mass result keeps the concept figure of 89 g	CGD-CAL-001 v0.3 section 10
4	The stud terminals: M6 male stud, 13 mm shoulder 6 mm tall, through-hole pins	The cable lugs and the link clamp on the shoulder; the shoulder height sets the link height	CGD-DDR-003 P6; BOM line 9
5	The power switch package height of 2.3 mm (TOLT), in the 1.5 mΩ class decided on 2026-10-02, and its price of about \$5.00 each	It sets the gap pad squeeze with 3 mm spacers (a different height needs a different spacer); the price sets most of the cost over the target	CGD-DDR-003 P2; BOM line 4
6	The gap pad: squeezes from 1.0 to 0.7 mm at a load the board fixings can apply, and insulates to at least 100 V	It is the only insulation between the live switch tops and the plate	CGD-DDR-003 P2; BOM line 17
7	The TVS diode and switch avalanche energy against about 1 J	Short-circuit turn-off energy at a 15 μs delay	CGD-CAL-001 section 4
8	A printable flame-retardant polycarbonate filament (the cover is opaque, decided 2026-10-02); a clear cover only if clear polycarbonate sheet with a V-0 rating at the cover's wall thickness can be bought and formed	The cover is drawn for printing	BOM line 13
9	The light pipe: 3 mm round, 19.6 mm below a flange about 6 mm across, and the hole it presses into (drawn 3.2 mm); the status light's package (PLCC-4, 3.5 x 2.8 x 1.9 mm)	The pipe's foot is drawn 0.5 mm above the light; a different length or hole needs the cover or pipe changed	BOM lines 13, 15 and 19

■ Value engineering

Value-engineering target: USD 140.00 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 159.15 (USD 19.15 over the target).

Main cost drivers (CGD-CAL-001 v0.6, section 11): the eight 1.5 mΩ-class charge and discharge MOSFETs (line 4, \$40.00, of which \$20.00 is the step up from the 2.5 mΩ parts decided on 2026-10-02), supporting electronics with the status light (line 15, \$16.35), the pack fuse and holder (line 8, \$12.00), the aluminium base plate, the main PCB and the secondary protector with its SCP fuse (lines 1, 2 and 14, \$10.00 each). The secondary protector and the parts added for construction (lines 17 and 18) account for \$14.00 of the estimate; the status light and light pipe for \$1.15.

Savings worth trying:

- Ask for a quantity price on the switches: power MOSFETs are usually markedly cheaper at quantity 100 or more, and each \$1.00 off the unit price saves \$8.00 on the board.
- Confirm the SCP fuse price, which may add to the estimate (CGD-DDR-003, A2).
- The status light stays (decided 2026-10-02) and the cover stays printed flame-retardant polycarbonate (decided 2026-10-02), so neither is a saving to try.

■ Decisions made

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	BQ76952-class front end; STM32G0B1-class controller building on the Libre Solar BMS firmware; BQ77216-class secondary protector with protector fuse; 40 A rating; high-side switching; LFP and NMC-capable hardware, LFP firmware first; SwapCell v0.3 CAN message set as a profile	Amish: "i accept all your recommendations, go with them across all repos."	CGD-DDR-001, CGD-DDR-002
2026-09-25	Budget raised from \$120 to \$140	Amish, same instruction	CGD-DDR-002
2026-09-25	Reused Apache-2.0 firmware kept in <code>firmware/third_party/</code> with its notices and listed in <code>LICENSE-SOFTWARE</code>	Amish, same instruction	CGD-DDR-002
2026-09-25	Protection settings (SCD 100 mV at 15 μ s, OCD2 40 mV at 20 ms, OCD1 24 mV at 320 ms, OCC 12 mV); top-side cooled TOLT switches; DPAK-class precharge switch; one-point current calibration at build	Amish, same instruction	CGD-DDR-002
2026-09-30	Design for construction: the changes P1 to P8 that make the board buildable	Made under Amish's 2026-09-30 instruction ("fix the design assumptions to match and be physically feasible"). The changes themselves were accepted on 2026-10-02 (below)	CGD-DDR-003
2026-10-02	Design for construction accepted: the changes P1 to P8 (flat plate with bought spacers and screws, squeezed gap pad, cover pillars and notches, fixing positions, fuse holder in line with B+ and copper link, stud terminals, mounting holes, connector position), as made	Amish: "i approve your recommendations for all 555 open decisions."	CGD-DDR-003, A1
2026-10-02	Cover material: printed flame-retardant polycarbonate, as drawn (option a); sheet aluminium is not used, since a metal cover would sit a few millimetres from live 58 V studs and nuts	Amish: "i approve your recommendations for all 555 open decisions."	CGD-DDR-001 item 12; CGD-DDR-002
2026-10-02	Cover colour: the prototype cover is opaque, with a light pipe over the status light; a clear cover only if clear polycarbonate sheet with a V-0 rating at the cover's wall thickness can be bought and formed	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26 render session, item 1
2026-10-02	R6 board loss: fit 1.5 m Ω -class switches in the TOLT package (option a), bringing the board loss to about 3.5 W; the protector fuse stays on the board	Amish: "i approve your recommendations for all 555 open decisions."	CGD-CAL-001 v0.3 section 12; CGD-DDR-002
2026-10-02	R8 state of charge: the firmware prompts for a full charge when about five days pass without one (option b), and R8's partial-cycling clause is restated to apply between prompted full charges; the 0.5 m Ω shunt is not fitted	Amish: "i approve your recommendations for all 555 open decisions."	CGD-CAL-001 v0.3 section 12; CGD-DDR-002

Date	Decision	Decided by	Record
2026-10-02	R14 mass: target relaxed to 0.7 kg (option b); the 4 mm plate is kept	Amish: "i approve your recommendations for all 555 open decisions."	CGD-CAL-001 v0.3 section 10; CGD-DDR-002; CGD-DDR-003
2026-10-02	State-of-charge check: cycle a 4S 20 Ah LFP pack (the worst case for R8) through seven days of partial cycles with a bench supply and a low-cost DC electronic load, count charge independently with a calibrated reference shunt and meter, then discharge fully to measure the true state of charge	Amish: "i approve your recommendations for all 555 open decisions."	CGD-DDR-001 item 13
2026-10-02	First test partner: a small off-grid solar installer that builds 16S LFP packs, testing on their bench rather than in a customer system (the first candidate to approach)	Amish: "i approve your recommendations for all 555 open decisions."	CGD-DDR-001 item 9
2026-10-02	Status light: one three-colour status light added to the supporting electronics, line 15 (option a)	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26 render session, item 4
2026-10-02	Product renders: the pack beside the board in the product renders only; engineering media keep the board above the pack	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26 render session, item 2
2026-10-02	Product renders keep the cover fixing and power cables, updated to the four pillars of CGD-DDR-003	Amish: "i approve your recommendations for all 555 open decisions."	Review note, 2026-09-26 render session, items 3 and 5
2026-10-02	SwapCell agreement to the NMC profile and the v0.3 CAN message set: record it in the SwapCell repo as well (Amish owns both), but only after the board envelope question is settled (SwapCell expects about 230 x 58 mm, 10 mm or less thick, at \$120; CellGuard is 220 x 110 x 33 mm at \$138)	Amish: "i approve your recommendations for all 555 open decisions."	CGD-PRC-001 open questions; review note cross-repo actions